

NAME:PRACTICE MIDTERM III

Please show your work. Calculators and notes are forbidden. Only 1 brain (your own) allowed per exam. Please also note that some of the problems have multiple parts.

0: (Thanks to Souvik Gouswami for this problem.)

(a) Consider the vector space P_3 , consisting of polynomials, in one variable, with degree ≤ 3 and real coefficients. Please determine whether the polynomial $x^3 + 6x + 4$ is in the span of $\{x^3 + x^2 + x, x^3 + 2x - 6, x^2 - 5\}$.

(b) Consider the vector space $\mathbb{R}^{2 \times 2}$, and the subspace W consisting of all matrices of the form $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where $a + b = c$, $b + c = d$, and $c + d = a$. What is the dimension of W ?

Hint: Both problems can be easily converted into questions about subspaces in $\mathbb{R}^4$.

1: Suppose a is any column vector in $\mathbb{R}^n$ **not** equal to the zero vector.

(a): Please show that $L(x) := \frac{aa^T}{a^T a}x$ is the map which projects any vector x orthogonally onto the line generated by a . In other words, please show that (1) $L(\alpha a) = \alpha a$ for all $\alpha \in \mathbb{R}$, and (2) $L(x) = 0$ if x is orthogonal to a .

(b): Let $R(x) := x - 2L(x)$. (R is known as a **Householder Reflection**, and is quite important in computing the QR factorization in a numerically stable manner.) Please show that R is the reflection which flips vectors across the hyperplane $H_a := \{x \in \mathbb{R}^n \mid a^T x = 0\}$. In other words, please show that (1) $R(x) = x$ for all $x \in H_a$, and (2) $R(\alpha a) = -\alpha a$ for all $\alpha \in \mathbb{R}$.

2: The following algorithm takes an input set of column vectors $a_1, \dots, a_n \in \mathbb{R}^n$ and produces a new set of column vectors $q_1, \dots, q_n \in \mathbb{R}^n$, along with a set of numbers $\{r_{ij}\}$:

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1. FOR  $i = 1$  TO  $n$ 
2.    $q_i = a_i$ 
3.   FOR  $j = 1$  TO  $i - 1$ 
4.      $r_{ji} = q_j^T q_i$ 
5.      $q_i = q_i - r_{ji} q_j$ 
6.   END FOR
7.    $r_{ii} = \|q_i\|$ 
8.   IF  $r_{ii} = 0$ 
9.     STOP
10.  END IF
11.   $q_i = q_i / r_{ii}$ 
12. END FOR
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(a): Please execute the above algorithm for the following inputs: $n = 3$, $a_1 := \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$, $a_2 := \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$, $a_3 := \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$.

(b): Please find the QR factorization of A .

3: Please find a basis for the subspace of $\mathbb{R}^4$ perpendicular to the span of $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$.

4: The following information is known about a matrix A :

$$A \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ and } A \begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

Also, every column of A is a multiple of $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

(a) What is the dimension of the right-nullspace of A ? Please explain why.

(b) What is the dimension of the row space of A ? Please explain why.

(c) What is the dimension of the left-nullspace of A ? Please explain why.

5: Suppose the following information is known about a matrix A :

$$A \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} = 9 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad A \begin{bmatrix} 1 \\ -3 \\ 9 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -3 \\ 9 \end{bmatrix}, \quad \text{and} \quad A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

(a): Please find the eigenvalues of A .

(b): Please find the eigenvectors of A .

(c): Is A invertible? Please explain...

(d): Is A diagonalizable? Please explain...

6: Find an orthonormal basis for the subspace of $\mathbb{R}^5$ generated by the following pair of vectors:

$$\begin{bmatrix} 1 \\ 3 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}.$$